SHWEATHA BATHINA MALLIKARJUNA

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New Brunswick, New Jersey

EDUCATION

Rutgers University, The State University of New Jersey — **New Brunswick, USA**

Master of Science in Computer Science | GPA: 3.77/4 — **Expected Graduation Dec 2026**

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks.

R.M.D. Engineering College — **Chennai, India**

Bachelor of Engineering in Computer Science | GPA: 9.24/10 — **Nov 2020 – June 2024**

Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Systems, Software Engineering.

EXPERIENCE

Rutgers Democracy Lab, Eagleton Institute of Politics — **New Brunswick, NJ**

Website Developer Intern — **Sept 2025 – Nov 2025**

- Improved performance and accessibility of a web platform serving 1000+ users, increasing user engagement and usability across academic content systems.
- Developed reusable blog and event modules using HTML, CSS, JavaScript, and MySQL, improving content management efficiency for non-technical users.
- Collaborated with cross-functional stakeholders to translate requirements into production-ready features.

Rutgers University — **New Brunswick, NJ**

TA – Grader, CS213 Software Methodology — **Sept 2025 – Jan 2026**

- Evaluated and debugged 100+ Java projects, focusing on JUnit test coverage, exception handling, and implementation of design patterns (Singleton, Factory, MVC).

Edu-Ideal Academy — **Chennai, India**

Computer Science Trainer — **June 2024 – Dec 2024**

- Delivered hands-on instruction in Python, C, and C++ to over 150 students, emphasizing problem-solving and algorithmic thinking.
- Mentored students through software development projects while tracking academic progress using cloud-based systems.

ACADEMIC PROJECTS

Role Ready – AI-Powered Resume & Job Analysis Platform | Python, RAG, LangGraph, Gemini, LangChain

- Built an evidence-grounded AI platform using RAG to retrieve relevant resume evidence and generate role-specific insights.
- Orchestrated multi-step AI workflows with LangGraph and Gemini, integrating retrieval, reasoning, and structured output generation.
- Designed prompt and output validation checks to improve response relevance, consistency, and grounding.

Face Recognition Attendance System | Python, OpenCV, Firebase

- Developed real-time face recognition achieving ~95% accuracy for 100+ students and automated attendance tracking.

Phishing Aware – Neuro-Fuzzy Anti-Phishing System | Java, JavaScript, MySQL

- Developed a neuro-fuzzy ML system to detect and block phishing URLs on fog networks, exploring AI-driven approaches to cybersecurity.

Feminine Impunity – Women Safety Application | Java, Firebase, Android

- Led development of an Android safety application with real-time location sharing, emergency alerts, and AI-assisted threat detection.

SKILLS

AI / Agent Engineering: Python · LLMs · RAG · LangGraph · LangChain · Prompt Engineering · NLP · Gemini · AI/ML · Vector Embeddings
Programming & Software: Python · Java · C++ · JavaScript · SQL · C · REST APIs · React · Spring Boot
Cloud & DevOps: Docker · Kubernetes · AWS · GCP · Git
Databases & Data: PostgreSQL · MySQL · MongoDB · Cassandra · Pandas · NumPy
ML / Deep Learning: TensorFlow · PyTorch · Scikit-Learn · Keras · OpenCV
Testing & QA: JUnit · Software Testing · Debugging · Test Coverage · Exception Handling · Edge-Case Testing · GitHub
Tools: Jupyter Notebooks · IntelliJ · PyCharm · Eclipse · Android Studio · Agile

RESEARCH, PUBLICATIONS & PRESENTATIONS

- Enhancing Supply Chain Management Efficiency through 4PL Integration: Leveraging Recent Technological Advancements — Presented at the 5th International Conference on Emerging Technology (INCET), Jain College of Engineering, Belagavi, May 2024; SCOPUS-indexed publication.

- Published research on ChatGPT, machine learning, and face recognition in IJIRCCE, IJMRSET, and IJIRSET; presented research on cybersecurity/steganography and CNN-based traffic sign detection at national and international conferences.